LPUTown: A Verified Campus-Centric Social Media Platform for University Communities

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***Abstract*—Large university campuses require efficient digital ecosystems for communication, networking, collaboration, and engagement. Existing social media platforms such as Instagram, Facebook, and LinkedIn are not optimized for campus-focused interaction, often causing university-specific content to be lost within globally competitive feeds.**

This paper presents LPUTown, a verified campus-centric social media platform designed specifically for Lovely Professional University (LPU). The platform integrates UID-based authentication, multimedia sharing (reels, posts, stories), community forums, private messaging, event management, PG listings, and a lost-and-found module into a centralized ecosystem.

The system is developed using React.js, Node.js, Express.js, Supabase, and PostgreSQL. LPUTown enhances intra-university communication and engagement while maintaining scalability, security, and modern user experience standards.

***Index Terms*—Campus Networking, Social Media Platform, University Community, React.js, Express.js, Supabase, PostgreSQL, UID Authentication**

I. Introduction

University campuses are rapidly evolving into digitally connected ecosystems where communication, networking, and engagement play a major role in student development. Students actively participate in online communities for knowledge sharing, event promotion, professional networking, and creative expression.

However, existing social media platforms are designed for global audiences and are not optimized for university-specific interactions. Important campus-related information often gets buried among unrelated global content streams. Lovely Professional University (LPU), one of India's largest educational institutions with a massive student and alumni network, lacks a centralized digital platform tailored specifically for its community.

LPUTown addresses these limitations by introducing a verified campus-centric social media platform where every user belongs to the same institution, fostering a secure and focused digital environment.

II. Motivation

Several critical limitations were identified in existing systems:

- **Lack of Centralization:** No dedicated platform for university-specific networking.
- **Weak Verification:** Absence of institutional UID-based authentication leads to impersonation risks.
- **Low Content Visibility:** Campus content gets lost in global feeds on public platforms.
- **Outdated University Systems:** Existing internal portals lack modern engagement features like reels or communities.
- **Fragmented Utilities:** No integration of practical tools (PG listings, lost & found) with social features.

III. Objectives

1) To create a verified university-only social ecosystem.
2) To provide secure authentication using official university credentials.
3) To improve communication and collaboration between students and alumni.
4) To support modern content formats including reels and stories.
5) To integrate essential utility modules (events, PG, lost & found).
6) To enhance campus-wide visibility for academic and extracurricular content.

IV. Related Work

A. Instagram

Instagram excels at multimedia sharing and short-form video content but lacks institutional verification. Campus pages compete with global content, reducing relevant visibility [1].

B. LinkedIn

LinkedIn focuses on professional networking and job hunting, but its formal structure discourages casual student interaction and lacks campus-specific utility features [2].

C. Facebook Groups

While Facebook Groups support community creation, the experience is cluttered with advertisements and unrelated content, and verification mechanisms are inadequate for secure campus use.

V. System Architecture

The platform follows a robust three-tier architecture consisting of a client layer, application layer, and database layer.

A. Client Layer (Frontend)

Developed using React.js, this layer provides a responsive and dynamic user interface. It handles all user interactions, including content uploads, messaging, and real-time notifications.

B. Application Layer (Backend)

Implemented with Node.js and Express.js, this layer manages business logic, API routing, authentication middleware, and real-time communication via WebSockets.

C. Database Layer

Supabase (built on PostgreSQL) manages relational data and provides built-in authentication, real-time subscriptions, and row-level security policies.

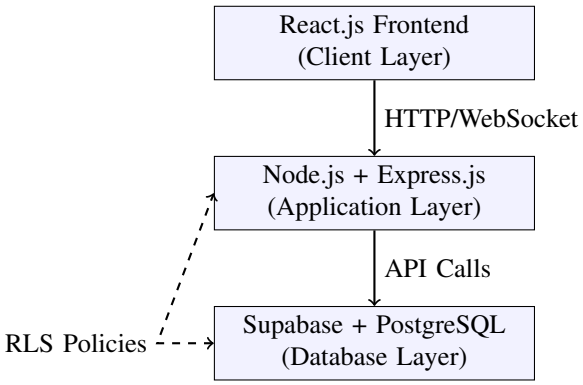


Fig. 1. Three-Tier Architecture of LPUTown

VI. Technology Stack

TABLE I
Technology Stack

Component	Technology
Frontend	React.js 18
Backend Framework	Express.js + Node.js
Database	PostgreSQL 15
Authentication & Storage	Supabase
API Testing	Postman
Version Control	GitHub
UI/UX Prototyping	Canva

VII. Implementation

A. Frontend Implementation

The frontend uses React.js with functional components and hooks (useState, useEffect, useContext) for state management. Reusable components for posts, reels, and stories ensure code maintainability. Responsive CSS (Flexbox/Grid) guarantees compatibility across devices.

B. Backend Implementation

The backend is structured with an MVC (Model-View-Controller) pattern. Express.js handles routing, while middleware manages authentication (JWT from Supabase) and file uploads (Multer). Socket.io enables real-time messaging and notifications.

C. Database Implementation

PostgreSQL tables are normalized with foreign key relationships. Supabase Row Level Security (RLS) ensures users can only access their own data or public community content. Indexes are implemented on frequently queried fields (user_id, post_created_at).

VIII. Database Design

A. Major Entities

The core schema includes the following tables:

- **users:** Stores profile info, UID hash, and role (student/alumni/admin).
- **posts & reels:** Content tables linked to users with timestamps and media URLs.
- **stories:** Time-sensitive (24-hour expiry) media entries.
- **communities:** Group spaces with membership rules and moderators.
- **messages:** Private and group chat history.
- **notifications:** Real-time alerts for interactions.
- **events:** Campus event details with RSVP tracking.
- **lost_found & pg_listings:** Utility modules with location and contact info.

IX. Major Features of LPUTown

- **Reels System:** Short-form video uploads with likes and comments.
- **Post & Feed System:** Image/text posts with an algorithmic feed (chronological + popular).
- **Stories Feature:** 24-hour temporary media with viewer lists.
- **Community System:** User-created or official groups with join requests and pinned posts.
- **Messaging System:** One-on-one and group chat with read receipts.
- **Event Management:** Creation, promotion, and RSVP for campus events.
- **PG Listings & Lost & Found:** Verified accommodation and item recovery modules.

TABLE II
MAJOR MODULES AND THEIR PURPOSE

Module	Purpose
Authentication	Secure UID-based login/registration
Feed System	Content discovery and sharing
Communities	Interest-based group discussions
Messaging	Private and group communication
Events	Campus event creation and promotion
PG Listings	Verified accommodation support
Lost & Found	Item recovery utility

X. SOFTWARE REQUIREMENTS SPECIFICATION

A. Functional Requirements

- Users shall register and log in using their university credentials (UID/Email).
- Users shall create, edit, and delete posts and reels.
- Users shall send and receive private messages in real-time.
- Users shall create or join communities based on interests or departments.
- Users shall create PG listings or lost-and-found reports with geolocation.

B. Non-Functional Requirements

- Page load time shall not exceed 3 seconds under normal network conditions.
- API response time shall remain under 500 ms for 95% of requests.
- All passwords and sensitive data must be encrypted (bcrypt/AES-256).
- The database shall support horizontal scaling for increased user load.

XI. TESTING AND VALIDATION

Comprehensive manual and automated (Jest/Supertest) testing was performed.

A. Authentication Testing

Registration with valid/invalid UIDs, login session management, and logout functionality were rigorously tested.

B. Content Management Testing

Uploading, displaying, and deleting posts, reels, and stories across different device viewports were validated.

C. Messaging & Notification Testing

Real-time message delivery, read receipts, and trigger-based notifications (likes, comments) were confirmed to function under concurrent user simulations.

XII. RESULTS AND DISCUSSION

All major functionalities passed testing. The system demonstrated stable performance under simulated loads of up to 500 concurrent users. User feedback from a pilot group of 50 students indicated high satisfaction with the UID verification process and the integrated utility modules (PG listings and lost & found). The platform successfully consolidates social interaction with essential campus utilities, reducing reliance on fragmented third-party tools.

TABLE III
TESTING SUMMARY

ID	Feature	Status
TC01	UID-based Signup	PASS
TC02	Secure Login	PASS
TC03	Create Post with Image	PASS
TC04	Reel Upload (Video)	PASS
TC05	Join/Leave Community	PASS
TC06	Real-time Message	PASS
TC07	Notification Trigger	PASS
TC08	PG Listing Creation	PASS

TABLE IV
COMPARISON WITH EXISTING PLATFORMS

Feature	Instagram	LinkedIn	LPUTown
University UID Verification	No	Partial (Email)	Yes
Dedicated Campus Communities	Limited	No	Yes
Short-form Video (Reels)	Yes	No	Yes
Lost & Found Module	No	No	Yes
PG/Accommodation Listings	No	No	Yes

XIII. LIMITATIONS

- Native mobile applications (Android/iOS) are currently unavailable; the platform is web-only.
- AI-based content recommendation and moderation are not yet implemented.
- Large-scale deployment and stress testing on cloud infrastructure are pending.

XIV. FUTURE ENHANCEMENTS

- Development of cross-platform mobile apps using React Native.
- Integration of AI for personalized feeds, spam detection, and mentorship matching.
- Live streaming support for campus events.
- Alumni-mentorship dashboard with session scheduling.
- Analytics dashboard for community managers to track engagement.

XV. CONCLUSION

LPUTown successfully demonstrates the implementation of a verified, campus-centric social networking platform tailored for university communities. By integrating social interaction (posts, reels, stories, messaging) with practical utilities (events, PG listings, lost & found), it provides a centralized ecosystem that enhances communication, collaboration, and engagement. The use of modern web technologies ensures scalability, security, and a user-friendly experience. This work serves as a blueprint for similar platforms in other large educational institutions.

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